

Fair Drop — Implementation Roadmap

Ordered phases. Each phase ends with green tests and a short entry in `docs/phase0-decisions.md` §6 if a decision changed. Test stack: **Vitest** + `@cloudflare/vitest-pool-workers` (Worker/DO/D1 in workerd), **Playwright** (frontend), **pytest** (simulator).

Phase	Objective	Components touched	Depends on	Tests required	Completion criteria
1 — Cloudflare foundation	Monorepo, Worker serving SPA + <code>/api/v1/health</code> , DO class bound, D1 bound, secrets, CI	pnpm workspace, <code>apps/worker</code> , <code>apps/web</code> (blank), <code>packages/shared</code> , <code>wrangler.jsonc</code> , GitHub Actions	—	health route test, DO ping test, D1 ping test	<code>wrangler deploy</code> works. <code>/api/v1/health</code> returns <code>{worker, do, d1: ok}</code> in prod. Free-tier limits re-verified and noted
2 — Database/event model	D1 migrations (all Part A tables), DO SQLite schema (Part B), seed FUTUREFEST 2026, sessions + users	<code>migrations/</code> , <code>persistence/</code> , <code>/events</code> , <code>/session</code>	1	migration up from empty, unique-constraint tests (<code>email_hash</code> , (<code>drop,user</code>)), session <code>create/recover/revoke</code> , cookie HMAC tamper test	GET <code>/events/futurefest-2026</code> live. Session survives refresh in browser. Revoked cookie → 401
3 — Drop state machine	<code>lifecycle/</code> with all transitions, guards, alarms, virtual clock, HALTED	DropDO, admin endpoints	2	table-driven test of every legal/illegal transition, alarm-driven open/close, virtual clock monotonicity	Drop goes DRAFT→... →CLOSED via clock on both WALL and VIRTUAL. Illegal transitions → 409 with no change
4 — Fair allocation engine	Registration, dedupe, freeze, snapshot hash, commit-reveal, <code>fairdrop-hmac-rank-v1</code> , waitlist	<code>admission/</code> , <code>allocation/</code> , <code>packages/shared/allocation.ts</code>	3	golden-vector test (fixed seed+snapshot → fixed winners), determinism across runs, statistical test (10k simulated lotteries: per-position win frequency $\chi^2 p > 0.01$), 50k-allocation CPU benchmark in workerd, repeat register → 1 participant	<code>lottery_entries == eligible</code> . Browser and server produce identical results. 50k allocation within DO CPU budget
5 — Inventory/reservation engine	Tickets, offers, holds, confirm, release, expiry (alarm/lazy/virtual), promotion, invariant checker	<code>inventory/</code>	4	property-based test (fast-check: random op sequences preserve I-3/I-4/I-5), concurrent-reserve test (1,000 parallel calls), expiry → promotion test, idempotent confirm	Zero violations across 10k random sequences. Invariant breach → HALTED (tested by fault injection)

Phase	Objective	Components touched	Depends on	Tests required	Completion criteria
6 — Request/identity controls	L0 RL binding, L1/L2 buckets, semantic + key idempotency, concurrent sessions, device binding, email normalization	Worker middleware, controls/	5	bucket math unit tests, 429 + Retry-After, Idempotency-Key conflict 422, token-reuse → revoke, normalization table tests	Flooding one session yields $\leq$ bucket allowance. Repeats never create state
7 — Behavioral abuse engine	Telemetry summary schema, browser aggregator, server features, risk-v1 scoring, Turnstile challenge flow, decisions persistence (on change)	risk/ , apps/web/src/telemetry , /telemetry , /challenge/verify	6	score engine unit tests incl. "single category can't exceed THROTTLE", accessibility cases (keyboard-only → ALLOW), Turnstile verify with Cloudflare test keys (always-pass/always-fail), payload $\leq$ 4 KB	Every non-ALLOW decision has reasons. Telemetry $\leq$ 1 req/10 s per tab
8 — Normal booking frontend	Routes 1–6 with consumer design system, rehydration from /me , polling contract, error states	apps/web	2–7	Playwright: full happy path, refresh at every step preserves state, offline/reconnect, waitlist + promotion path, challenge path	A human can complete register → win → hold → confirm on the deployed site, surviving refresh at each step
9 — Live WebSocket/dashboard layer	DO hibernation hub, 500 ms snapshots, sampled event feed, viewer tokens	live/ , metrics/ , /experiments/:id/live	3, 5	WS auth tests, broadcast cadence, viewer cap, reconnect resumes with latest snapshot	Dashboard client receives snapshots under load without increasing DO request count beyond budget
10 — Local stress simulator	Python package: population generator, profiles, virtual-clock timeline engine, logical + real HTTP drivers, client latency histograms, independent metric recomputation	simulator/ (uv, httpx, pydantic)	1 (API stubs), contract	pytest: deterministic population for seed, population_hash stability, profile parameter distributions, timeline ordering, offline dry-run against mock server	fairdrop-sim plan --seed 42 reproducible. Dry-run produces expected op counts
11 — Controlled simulation API	/sim/experiments , batches (exactly-once batch_seq), control, client-metrics, finalize, labels, capacity model in DO, experiment drops, sim tokens, virtual	Worker, DropDO simBatch , D1 experiments/metrics/artifacts	4–7, 9, 10	auth negative tests (no token, wrong scope, sim token on public drop → 403, header spoof ignored on public drop), batch replay idempotency, out-of-order 409, labels hash mismatch 422	50k logical FAIR run completes end-to-end on deployed infra within budget. Results computed server-side match

Phase	Objective	Components touched	Depends on	Tests required	Completion criteria
	network header guard				simulator recomputation
12 — Stress-test UI	/lab (admin login, scenario builder, profile mix sliders, seeds, launch command generator/run status), /lab/experiments/:id/live dashboard	apps/web console design system	9, 11	Playwright: create experiment, view live dashboard with a running sim, "planned vs labelled" display	A judge can watch a live run with all required dashboard fields
13 — Results + audit	Results page, comparison page, audit record finalization, snapshot artifacts, /audit/verify, in-browser verification	D1 audit/artifacts, apps/web results + audit	4, 11	audit tamper tests (modified snapshot/seed → verify fails), browser recompute equals server, results render "—" for missing data	Audit page verifies a 50k drop in-browser. All fairness metrics shown with CIs
14 — Naive baseline	naive-fcfs-v1 mode, paired comparison runs, comparison view	DropDO mode switch, simulator compare command	5, 11, 13	same population hash for both modes, naive integrity also zero, deterministic naive outcome in logical mode	fairdrop-sim compare --scenario mixed attack --seed 7 produces a comparison page with real numbers for both modes
15 — Adversarial hardening	Run all scenarios + amplification sweep + ablations + race_reserve. Fix findings. Tune thresholds against human false positives. Optional drand (OD-4)	all	14	full scenario matrix, regression suite of previously found bugs, human FP rate ≤ target (e.g. ≤ 2 % challenged)	All integrity metrics 0 in all runs. Findings documented. Residual risks updated in threat model
16 — Deployment & demo prep	Production config, custom domain, reset/seed scripts, demo runbook, pre-computed <i>real</i> comparison runs stored for backup, quota check	ops scripts, README, docs/demo-runbook.md	15	smoke test script against prod, quota headroom check	Dry-run demo end-to-end twice on prod. Runbook includes fallback if quota is near limit

Dependency graph

Parallelizable after Phase 3: frontend scaffolding (8) and simulator (10) can proceed against the API contract with mocks.